

CanopyIQ

An AI-Assisted Decision Support System for Urban Tree Management and Community Prioritization

CSCN8040 - Case Studies in AI and ML | Final Team Presentation

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decision support, not automation

live demo included



The city problem is not “more AI.”

It is how to defend difficult forestry decisions with limited time and budget.

A planner has to answer three questions:

- Which communities need action first?
- Why this area before another?
- Can we explain the recommendation to council and residents?

CanopyIQ turns scattered tree records into an explainable priority list.



Who is affected?

Canopy decisions shape budgets, work orders, resident trust, and climate adaptation.

Municipal planners

Need defensible rankings

Arborists & crews

Need field-review queues

Residents

Experience shade, service, and risk

Council

Needs a public explanation

The system must be explainable enough for public decision-making, not just accurate in a notebook.

Research question and hypothesis

The final report focuses on evidence we can defend.

Research question

How can open municipal tree inventory and environmental data be integrated into a reproducible model to identify high-priority areas for urban forestry intervention?

Hypothesis

Combining tree asset condition, canopy context, and operational indicators improves resource prioritization and makes forestry decisions easier to explain.

Final research hierarchy

Primary: transparent community index

Validation: baselines + Monte Carlo stability

Secondary: observed removal classifier

History: synthetic/budget/GenAI prototypes

Data used in the final system

We combined municipal inventory records, planning boundaries, Census context, and real LST cache data.

152,233

raw municipal tree records

145,326

harmonized classifier records

55

planning boundary polygons

51

rankable communities

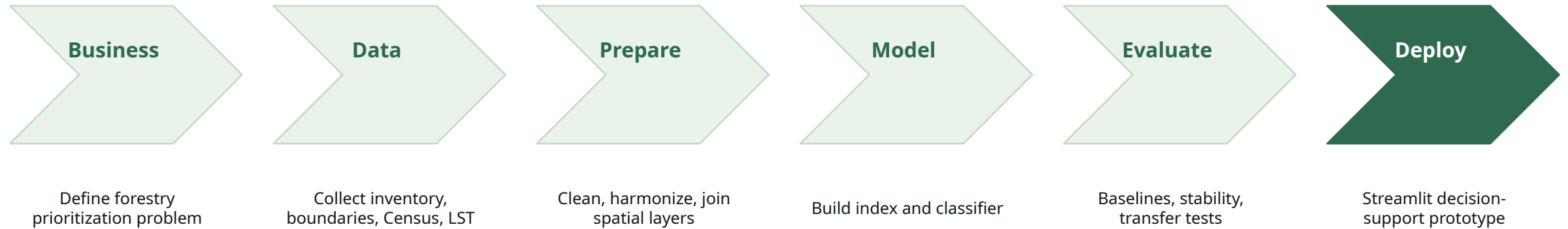


Source layers

- Kitchener Tree Inventory
- Waterloo Street Tree Inventory
- Kitchener planning communities
- 2021 Census population/income
- Landsat/GEE summer land surface temperature

Methodology: from raw records to decision support

CRISP-DM helped keep the work connected to the municipal problem.



Core implementation

Python notebooks for analysis

GeoPandas/Shapely for geospatial joins

scikit-learn for evaluation

Streamlit app for live demonstration

The CanopyIQ Index is intentionally transparent

Each community score is an equal-weighted average of four explainable indicators.

$$\text{Score} = 0.25D + 0.25A + 0.25M + 0.25B$$

D

Density deficit

Where living tree density is
low

A

Attrition

Where records show
higher removal/loss

M

Monoculture

Where genus diversity is
lower

B

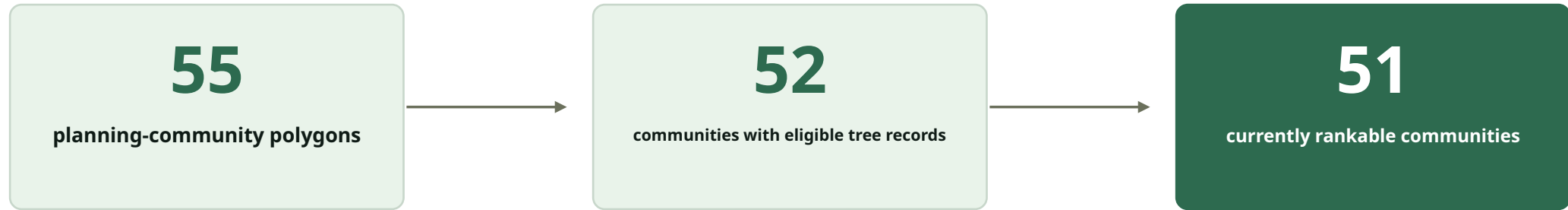
Backlog

Where inspection recency
is weaker

Equal weights are a starting point, not a claim of neutrality. The final system should support stakeholder weight review.

Data integrity check: 55 boundaries became 51 scores

We explain the difference instead of hiding it.



Not represented by eligible tree records: Dundee, Pioneer Tower East, Victoria North

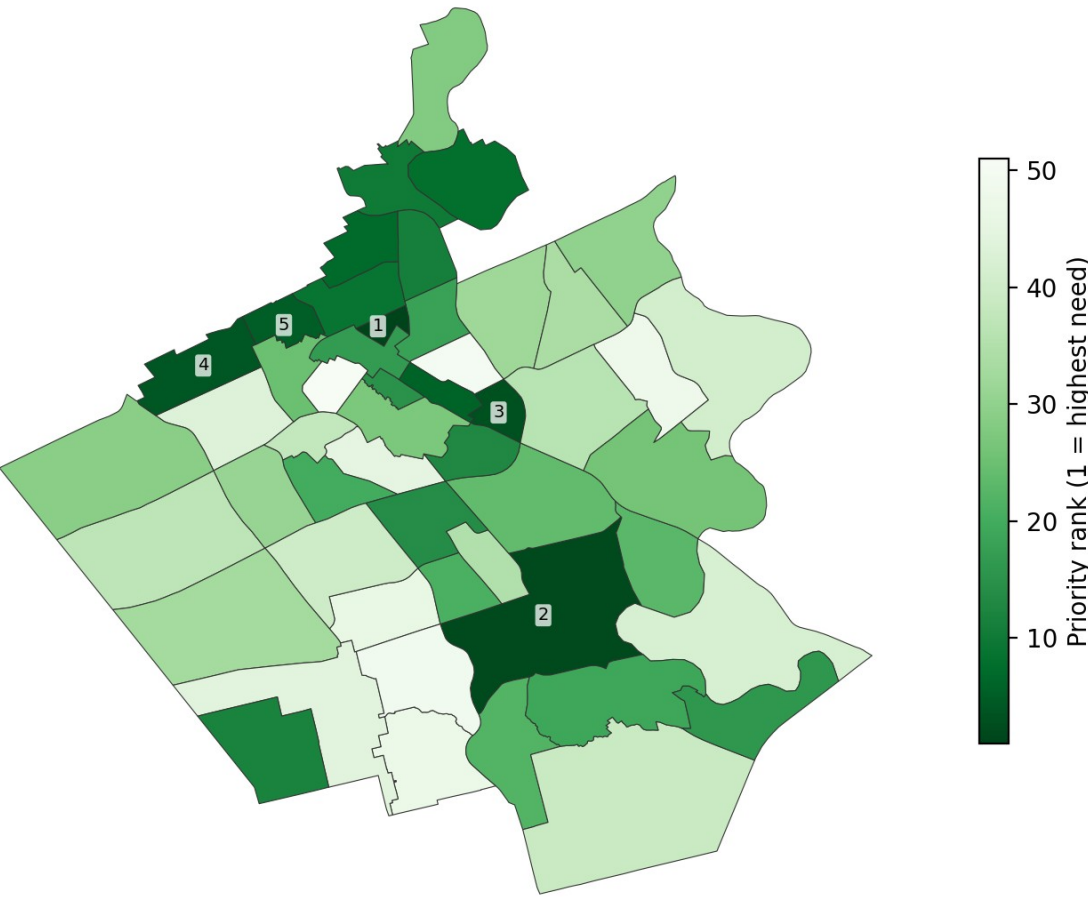
Excluded from final scoring because backlog was missing: South Plains

Result: final ranking is honest about missing data and scope.

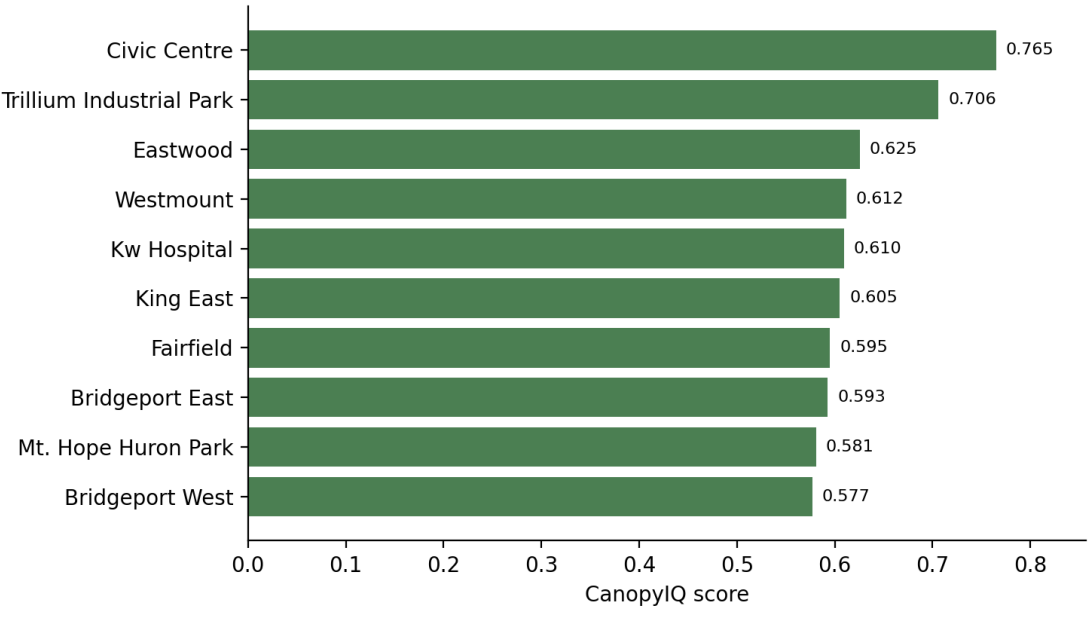
Result 1: community prioritization map

Civic Centre is rank #1 in the current scoreable output.

CanopyIQ Community Priority Index - Kitchener



Top 10 High-Need Communities by CanopyIQ Score



Civic Centre

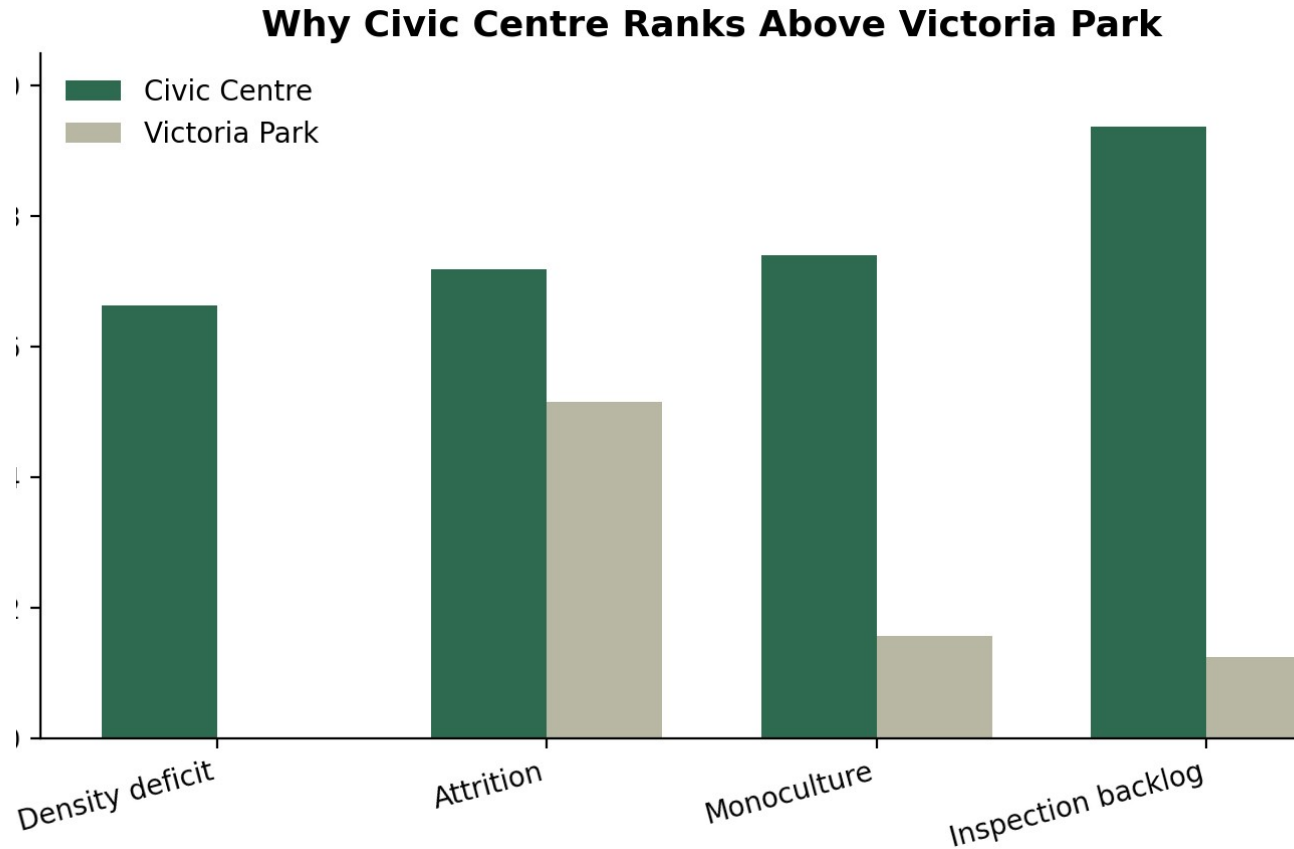
highest-need community

0.765

CanopyIQ score

Result 2: the score is explainable

The app can show why one community ranks above another.



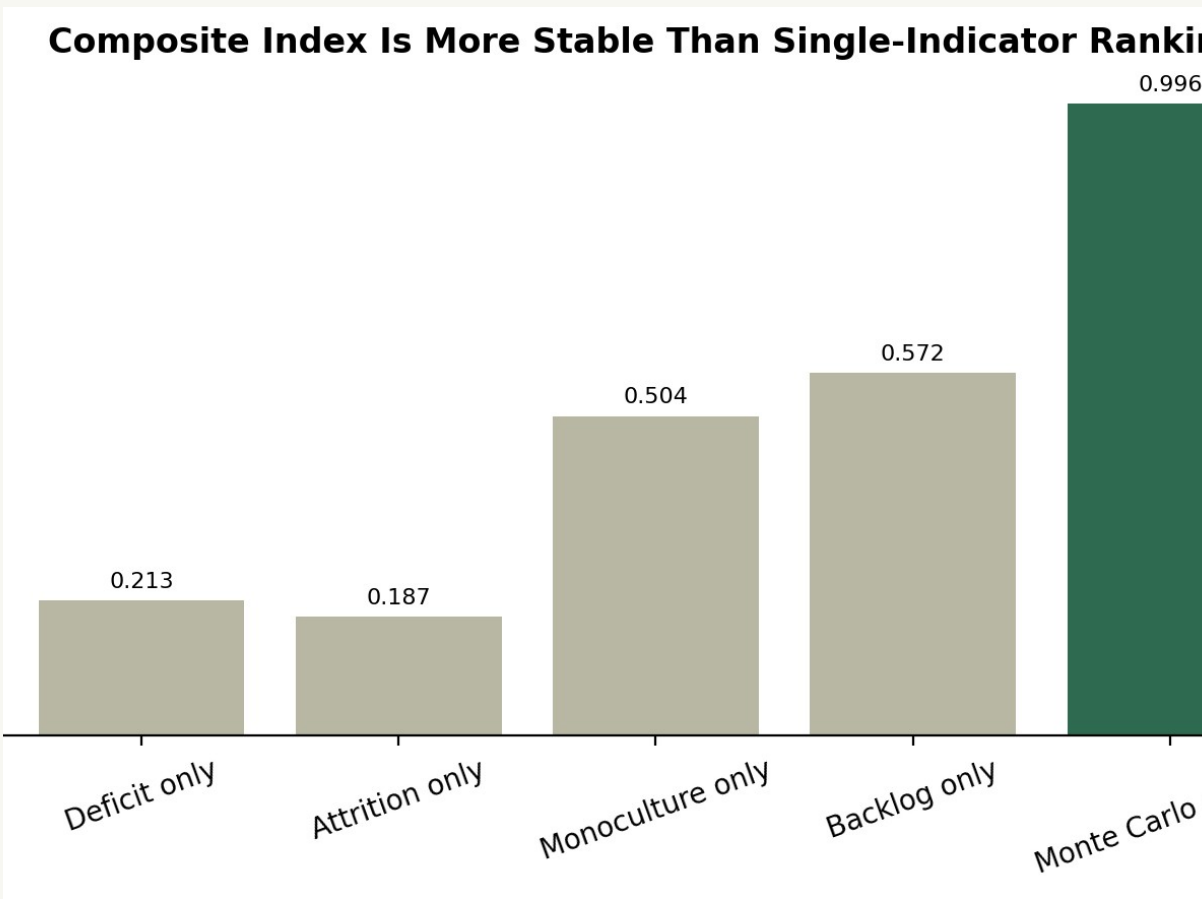
Civic Centre ranks high because it combines:

- Long inspection backlog
- High attrition signal
- High monoculture risk
- Lower tree density than many communities

That creates a defensible explanation, not just a number.

Result 3: the index is stable under weight changes

Monte Carlo testing reduces the risk that rankings are fragile.



1,000
Monte Carlo draws

0.996
median Spearman rho

0.990
5th percentile rho

100%
median top-15 preserved

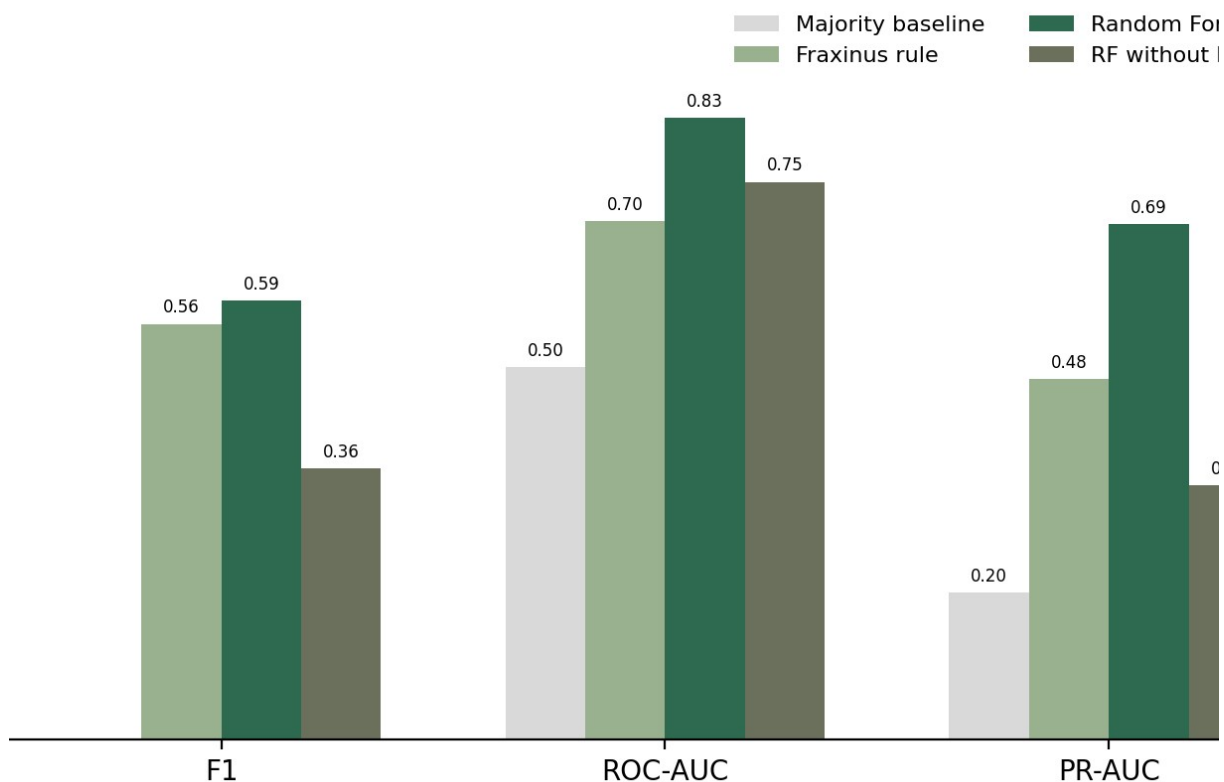
Interpretation

Reasonable weight perturbations did not materially change the high-need group. The index is transparent and stable, but weights should still be reviewed by stakeholders.

Secondary ML experiment: Random Forest improves over baselines

But it remains a planning signal, not an autonomous decision.

Tree Removal Classifier: Baseline vs Random Forest



0.589

Pooled RF F1

0.835

Pooled RF ROC-AUC

0.692

Pooled RF PR-AUC

0.558

Fraxinus baseline F1

The Random Forest beats the simple Fraxinus rule on F1 and ranking quality.

This is useful, but not enough to authorize removals without field inspection.

What the classifier taught us

It learns useful patterns, but those patterns are not universally portable.

No-Fraxinus test

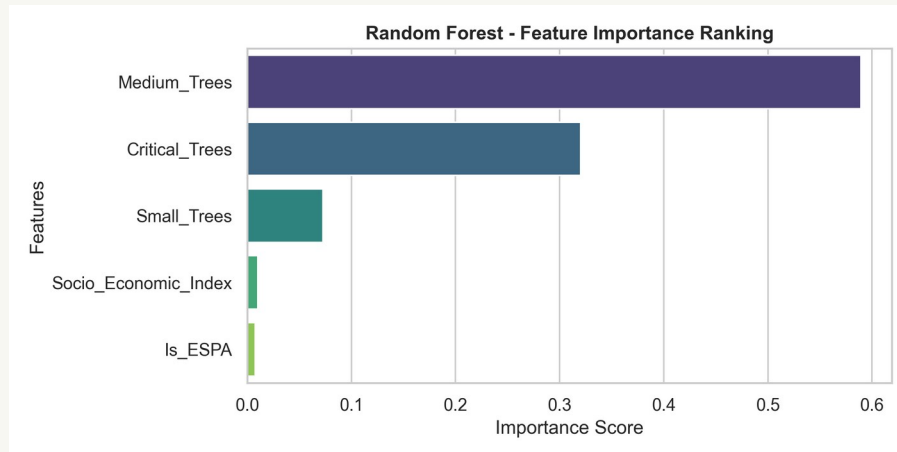
ROC-AUC drops to 0.749 and PR-AUC to 0.341.

Kitchener -> Waterloo

Transfer ROC-AUC = 0.773.

Waterloo -> Kitchener

Transfer ROC-AUC = 0.664.



Conclusion

The model is useful for prioritizing review. It is not a general tree-mortality oracle. Species history and city-specific data practices still matter.

Live demonstration: CanopyIQ prototype

Two minutes: map, community score, comparison, metrics, responsible AI.

1. Open Overview

2. Community Prioritization

3. Select Civic Centre

4. Compare with Victoria Park

5. Model Evaluation

6. Responsible AI



Demo rule: no fake production claims. This is a working decision-support prototype.

Responsible AI design controls

The strongest governance principle is simple.

CanopyIQ advises. Arborists decide.

Accessibility

Table equivalent for map outputs

Privacy

No inexact joins to individual tree coordinates

Security

Schema validation, RBAC, audit logs

Equity

Data coverage flag + Green Equity KPI

Human oversight

Certified arborist inspection before removals

Monitoring

Drift checks and annual retraining

Limitations we do not hide

These are practical, specific, and fixable with better data and field feedback.

Incomplete inventory coverage

Under-documented areas may appear lower need.

Missing indicator issue

South Plains was excluded because backlog was missing.

Value-laden weights

Equal weights need stakeholder review.

Species/pest dependency

Fraxinus/EAB history affects the classifier.

Cross-city mismatch

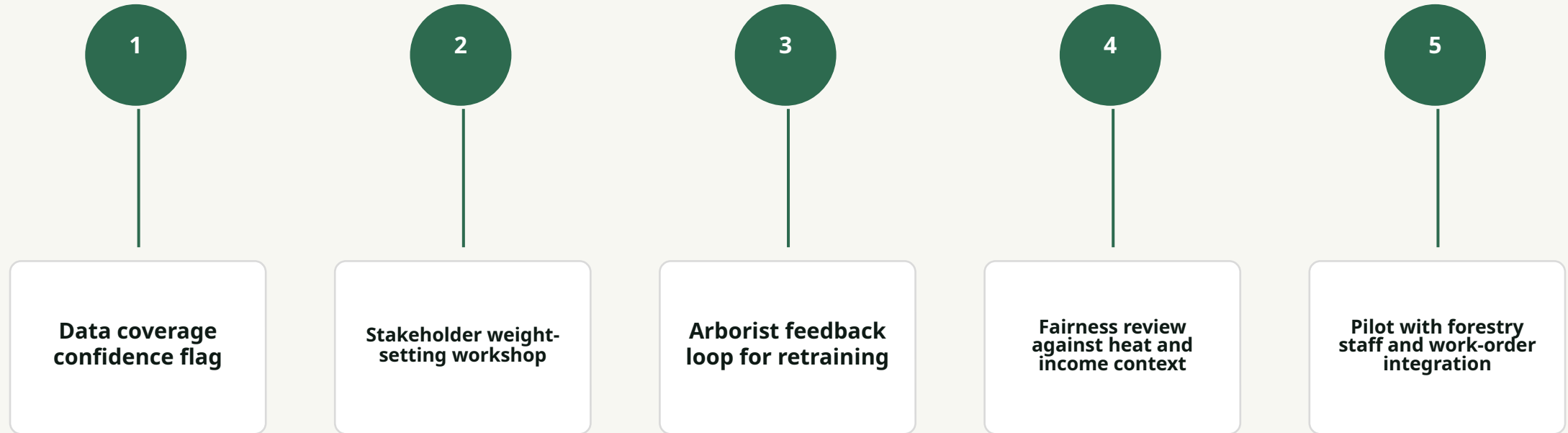
Transfer performance is asymmetric.

Prototype scope

No live Cityworks or AWS integration yet.

Future work: from prototype to pilot

The next version should improve data confidence, feedback, and municipal workflow fit.



Goal: keep the system explainable while making it more useful in real forestry operations.

Key takeaway

CanopyIQ makes urban forestry prioritization more explainable, but it must remain decision support.

Transparent index. Verified baselines. Working demo. Human accountability.

questions

